An AI-Driven Hybrid Smart Eco-Greenhouse System for Precision Lavender Cultivation and Premium Essential Oil Production

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***Abstract*— Lavender is a high-value aromatic crop whose essential oil yield depends critically on precise management of disease, microclimate, nitrogen nutrition, and lighting. Existing greenhouse systems treat these factors independently using generic rule-based controllers, leading to suboptimal yield and resource inefficiency. This paper presents an integrated AI-powered greenhouse management system unifying real-time disease detection, climate control, nitrogen stress diagnosis, and adaptive lighting into a single low-cost IoT architecture. The system employs YOLOv8m for real-time pest and disease detection, a hybrid Random Forest controller for climate regulation, MobileNetV3-Large for nitrogen stress classification, and VGG16 for growth-stage adaptive lighting. All components share a unified Flask REST backend with Firebase Firestore persistence and a Flutter cross-platform user interface. Comprehensive experimental evaluation demonstrates 86.4% nitrogen stress classification accuracy, mAP@0.5 of 0.899 for disease and insect detection, 38% reduction in lighting energy consumption, and 18.6% water reduction versus fixed-schedule irrigation. A 5-fold cross-validation confirms consistent model performance (mean accuracy 86.1 ± 1.4%). Total hardware cost is under $40 per node, making precision lavender cultivation accessible to small-scale farmers globally.**

***Index Terms*— Precision Agriculture, Lavender Cultivation, Deep Learning, YOLOv8m, MobileNetV3, Internet of Things, Nitrogen Stress Detection, Adaptive Lighting, Greenhouse Automation, Essential Oil Production.**

I. Introduction

Lavender (*Lavandula angustifolia*) is a high-value aromatic crop cultivated globally for essential oils used in cosmetics, pharmaceuticals, and aromatherapy. The global lavender oil market was valued at USD 218 million in 2022 and is projected to grow at a compound annual growth rate (CAGR) of 5.6% through 2030, driven by increasing demand in fragrance and therapeutic applications. Unlike conventional agricultural crops, lavender exhibits extreme sensitivity to environmental stressors. Suboptimal disease management, improper climate regulation, nitrogen imbalance, and inadequate lighting each independently reduce essential oil yield and commercial viability.

Small-scale farmers face a particular challenge: commercial greenhouse automation systems from vendors such as Priva, Ridder, and Growlink are prohibitively expensive, with installation costs exceeding $10,000 per greenhouse unit. Affordable solutions rely on generic, rule-based, single-function controllers that cannot address lavender's unique physiological requirements. This affordability gap represents both a market failure and a research opportunity: the convergence of low-cost microcontrollers (e.g., ESP32), open-source AI frameworks (YOLOv8, TensorFlow), and cloud services (Firebase) enables the construction of intelligent greenhouse systems at commodity hardware prices.

A. Problem Statement

Existing greenhouse management approaches suffer from four fundamental limitations when applied to lavender cultivation.

First, plant disease and pest detection systems are predominantly offline or desktop-based, requiring manual

image uploading and lacking real-time IoT integration. Most systems cannot simultaneously detect both disease and insect presence in live camera streams, nor can they trigger immediate hardware alerts such as buzzers or LEDs. Existing models are typically trained on generic datasets and are not fine-tuned for lavender-specific pathological signatures, which include subtle discoloration patterns and early-stage leaf symptoms that are visually similar to healthy states. Conventional detection systems also lack stability buffering, making them susceptible to false positives from single-frame noise, which causes unnecessary pesticide application and increased production costs.

Second, climate control systems predominantly rely on rule-based or generic automation approaches that lack crop-specific adaptability, real-time intelligence, and integrated control mechanisms. Most existing systems are designed for generic greenhouse crops using fixed thresholds or simple IF-ELSE rules that cannot adapt to dynamic environmental fluctuations or to the diurnal variations in temperature and humidity typical of enclosed growing spaces. Lavender requires moderate temperatures (22–28°C) and controlled humidity (55–75% RH), yet no existing affordable system is specifically tailored to these requirements. Conventional controllers operate fans and humidifiers on binary ON-OFF schedules or hysteresis bands, resulting in temperature overshoot, humidity cycling, and unnecessary energy consumption.

Third, commercial nutrient sensors measure electrical conductivity as a proxy for nitrogen concentration, but this indirect measurement is fundamentally inaccurate in greenhouse conditions, where fluctuating temperature and humidity cause significant reading drift. Visual assessment of nitrogen status is particularly difficult for non-experts, as deficiency (chlorosis) and lockout (yellowing from overwatering) present identical visual symptoms despite requiring diametrically opposite management responses: the former requires increased fertilization while the latter requires immediate irrigation reduction. Conventional irrigation follows fixed schedules, leading to overwatering, nutrient leaching, and delayed response to nitrogen toxicity events.

Fourth, greenhouse lighting systems use fixed timers or static spectrums. They do not adapt to real-time plant health or growth stage, wasting energy and failing to optimize essential oil synthesis. Lavender's photomorphogenic response is wavelength-dependent: blue light (450 nm) promotes vegetative growth and structural strength, while red light (660 nm) triggers flowering and essential oil biosynthesis. Providing the wrong spectrum at any growth stage reduces terpene yield by up to 34% according to controlled studies. No current solution integrates computer vision to detect growth stage or health status and adjust lighting profiles accordingly in real time.

No existing affordable system addresses all four limitations within a single, integrated architecture.

B. Research Contributions

This paper makes the following seven original contributions:

1) A unified AI-powered greenhouse management system that integrates disease/pest detection, climate control, nutrient and irrigation management, and adaptive lighting within a single architecture sharing a common Flask REST backend, Firebase Firestore database, and Flutter cross-platform user interface.
2) A YOLOv8m-based real-time detector for lavender disease and insects using ESP32-CAM live streams, achieving mAP@0.5 of 0.899, featuring 10-frame stability buffering that reduces false positives by 73%, and UDP-triggered LED/buzzer alerts with sub-5 ms latency.
3) A hybrid Random Forest climate controller combining regression (fan speed 0–100%, R^2=0.943) and classification (humidifier level 0–3, accuracy 91.3%) that maintains lavender-specific targets through closed-loop feedback from DHT22 and DS18B20 sensors with previous actuator states as feedback features.
4) A MobileNetV3-Large convolutional neural network classifying five nitrogen stress states with 86.4% accuracy through two-phase transfer learning and a hybrid vision-sensor approach that improves deficiency-versus-lockout discrimination by 16.3 percentage points over vision-alone baselines.
5) A VGG16-based adaptive lighting controller classifying four plant growth stages to adjust RGB LED spectra with LDR-guided PWM dimming, achieving 38% lighting energy reduction versus fixed-schedule operation.
6) A complete prototype with total hardware cost under $40 per greenhouse node, validated through 48-hour continuous operation testing, making precision lavender cultivation economically accessible to small-scale farmers.
7) Comprehensive experimental evaluation including 5-fold cross-validation, ablation studies quantifying the contribution of each system component, and statistical significance testing confirming the validity of reported improvements.

C. Paper Organization

Section II reviews related work. Section III presents the four-layer system architecture. Section IV specifies system requirements. Section V describes hardware design. Section VI details AI model design and training. Section VII explains the software implementation. Section VIII presents experimental results. Section IX discusses findings, limitations, and deployment recommendations. Section X outlines future work. Section XI concludes the paper.

II. Literature Survey

A. Plant Disease and Pest Detection

Deep learning, particularly CNNs, has enabled automated plant disease detection from leaf images. Mohanty et al. [1] trained a CNN on 54,306 images from the PlantVillage dataset, achieving 99.35% accuracy under controlled laboratory conditions but with significantly reduced performance in real-field settings. Ferentinos [2] confirmed this gap across multiple crop types. Lightweight attention networks [3] have improved robustness in complex visual backgrounds. Integration of YOLOv8 with ESP32-CAM for live monitoring has been explored [4], though multi-spectral analysis is often required for detecting subtle structural features [5]. No existing system provides simultaneous disease and insect detection with hardware-level alerts and detection stability buffering.

B. Greenhouse Climate Control

IoT-based climate control using Arduino, ESP8266, and ESP32 microcontrollers has been extensively studied [6]. Simple hysteresis-based rule controllers cause temperature overshoot and energy inefficiency [7]. PID improvements require manual tuning and struggle with nonlinear temperature-humidity coupling [8], [9]. ML-based prediction of greenhouse micro-environments has been studied [10], yet few systems integrate ML-driven closed-loop actuator control [11]. No crop-specific system tailored to lavender's temperature and humidity requirements with closed-loop state-aware control has been reported.

C. Nutrient and Irrigation Management

Nitrogen drives vegetative growth as the core component of chlorophyll. For lavender, the optimal nitrogen window is exceptionally narrow, with N-Burn causing complete loss of essential oil concentration. Laboratory soil testing provides accuracy but lacks the continuous temporal resolution needed for real-time intervention [12]. Commercial EC-based NPK sensors cannot chemically isolate nitrogen and drift under fluctuating environmental conditions [13], [14]. CNN-based nitrogen deficiency detection has been applied to rice [15], but N-Burn toxicity is largely ignored, and vision-only systems cannot discriminate between N-deficiency and N-lockout without supplementary sensor data [16].

D. Adaptive Lighting Systems

LED spectral composition directly influences plant terpene biosynthesis. Dynamic spectral control in CEA environments has been identified as a critical emerging technology [17]. Commercial greenhouse lighting uses fixed-spectrum LEDs or daily timers [18] with no adaptation to real-time plant status. Plant growth responds dynamically to spectral shifts [19], yet computer vision-driven spectral adjustment remains unreported. Adaptive dimming when natural sunlight suffices is rarely implemented [20], representing a significant energy efficiency gap.

TABLE I

Literature Gaps Motivating the Proposed System

Domain	Gap	Proposed Solution
Disease detection	No real-time IoT camera + hardware actuation	YOLOv8m + UDP alerts
Climate control	No crop-specific closed-loop state-aware control	Random Forest + feedback
Nitrogen diagnosis	No affordable N-Burn + lockout discrimination	MobileNetV3 + soil sensor
Irrigation	Fixed schedules, no adaptive nitrogen response	Adaptive flush protocol
Lighting	No vision-driven spectral adaptation	VGG16 + LDR dimming

TABLE II

Four-Layer Architecture: Responsibilities and Components

Layer	Responsibility	Key Components
Data Acquisition	Sensing, imaging, streaming	ESP32-CAM, DHT22, DS18B20, LDR, Capacitive sensor
Intelligence	AI inference, decision-making	YOLOv8m, Random Forest, MobileNetV3, VGG16
Control	Actuator command dispatch	Fan, humidifier, pump relay, RGB LED, buzzer
Application	User interaction, persistence	Flutter app, Flask API, Firebase Firestore

E. Summary of Research Gaps

Table I summarizes the key gaps identified across the reviewed literature. No existing system simultaneously addresses all five gaps in a single low-cost integrated architecture.

III. System Architecture

The proposed system follows a four-layer architecture: data acquisition, intelligence, control, and application. Table II summarizes the responsibilities and key components of each layer.

A. Data Acquisition Layer

The data acquisition layer comprises an ESP32-based node equipped with an OV2640 camera and multiple sensors. The camera captures live MJPEG streams at QVGA resolution (320×240 pixels) for disease detection, nitrogen stress diagnosis, and growth stage classification. Environmental sensors include a DHT22 for air temperature and humidity, a DS18B20 for soil temperature, a capacitive soil moisture sensor, and an LDR for ambient light intensity. Sensor data are transmitted to the backend server via HTTP at configurable intervals. Camera streams are served over HTTP on port 81.

The sampling strategy uses adaptive polling: for slowly varying parameters (soil temperature, soil moisture), a 10-minute polling interval is used; for fast-varying parameters (air temperature, humidity), a 30-second interval is used; and for disease detection, continuous streaming at 5 frames per second is maintained.

B. Intelligence Layer

YOLOv8m performs real-time object detection outputting bounding boxes with class labels (Lavender_Disease, Lavender_Healthy, Insect) and confidence scores. A 10-frame sliding window buffer calculates detection stability ratios to prevent false positives before triggering hardware alerts.

Random Forest models control the greenhouse climate. A regressor predicts optimal fan speed (0–100%) and a classifier predicts humidifier level (0–3: Off, Low, Medium, High). Both incorporate previous actuator states as feedback inputs, enabling closed-loop control without explicit system identification.

MobileNetV3-Large classifies nitrogen stress from leaf images across five classes: Healthy, N-Deficiency, N-Lockout, N-Burn, and Other Stress. Soil moisture data and fertilization history cross-validate the visual diagnosis to resolve the symptom ambiguity between deficiency and lockout.

VGG16 classifies plants into four growth states: Dormant, Vegetative (blue-heavy), Flowering (red-heavy), and Recovery (full spectrum). LDR sensor data enable PWM-based dimming when natural sunlight suffices.

C. Control Layer

For **disease and insect detection**, the flash LED (GPIO 4) and buzzer (GPIO 13) are triggered via UDP when the detection ratio exceeds a 30% threshold over 10 frames. A safety auto-timeout stops the LED after 30 seconds to prevent thermal damage.

For **climate control**, fan speed is regulated via PWM (0–100%) and humidifier level via relay (0–3). All actuators support Auto (AI-driven), Manual, and Off operational modes, enabling farmer override at any time.

For **irrigation**, a 12 V DC pump follows a baseline schedule under normal operation, pauses completely when N-Lockout is detected, and executes a Fresh Water Flush protocol (twice normal duration with clear water) when N-Burn is confirmed. The system enters a hold state awaiting user confirmation before resuming normal irrigation.

For **lighting**, RGB LED strips receive PWM signals to adjust both spectrum and intensity. LEDs are turned off when ambient LDR readings exceed the photosynthetically active radiation (PAR) threshold (>300 lux) to conserve energy without sacrificing plant growth.

D. Application Layer

A cross-platform Flutter application provides real-time sensor dashboards, live camera feeds with overlaid bounding boxes, manual actuator control panels, historical data visualization, PDF report generation, and push notifications. Data persistence is handled by Firebase Firestore, with the Flutter application synchronizing in real time using Firebase SDK listeners.

TABLE III
INTER-COMPONENT DATA FLOW SUMMARY

Source	Dest.	Data / Protocol	Frequency
ESP32-CAM	Flask	MJPEG stream / HTTP port 81	5 fps continuous
ESP32	Flask	JSON sensor data / HTTP POST	30 s / 10 min
Flask	ESP32	UDP command string / UDP 82	On-demand
Flask	ESP32	HTTP GPIO control / HTTP	On-demand
Flask	Firestore	Sensor & detection logs / SDK	Per event
Flutter	Flask	REST API request / HTTP	On-demand
Firestore	Flutter	Real-time snapshot / SDK	On change

TABLE IV
FUNCTIONAL REQUIREMENTS BY SUBSYSTEM

Subsystem	Requirement	Priority
Disease Detection	Detect disease, insect, and healthy states from live MJPEG stream	High
Disease Detection	Buffer 10 frames and alert only when ratio exceeds 30%	High
Disease Detection	Trigger LED and buzzer via UDP within 5 ms of threshold breach	High
Climate Control	Measure air temperature and humidity every 30 s	High
Climate Control	Predict fan speed and humidifier level using ML model	High
Climate Control	Support Auto/Manual/Off modes per actuator	Medium
Nitrogen/Irrigation	Classify leaf image into 5 nitrogen stress classes	High
Nitrogen/Irrigation	Cross-validate with soil moisture and fertilization history	High
Nitrogen/Irrigation	Execute flush protocol on N-Burn detection	High
Adaptive Lighting	Classify plant growth stage into 4 categories	High
Adaptive Lighting	Adjust RGB LED spectrum and PWM intensity per stage	High
Adaptive Lighting	Dim or disable LEDs when ambient LDR exceeds PAR threshold	Medium
Application	Display real-time sensor readings on cross-platform dashboard	High
Application	Log all sensor readings and events to cloud storage	Medium
Application	Generate and export PDF reports on demand	Low

E. Data Flow and Communication

Table III describes the data flow between system components, including protocol, direction, data type, and update frequency.

IV. System Requirements

A. Functional Requirements

Table IV documents the functional requirements of the system, organized by subsystem with priority classification.

B. Non-Functional Requirements

Table V documents the non-functional requirements with associated target metrics and test methods.

V. Hardware Design

A. Integrated Hardware Subsystem Overview

The proposed smart lavender farming system comprises four tightly integrated hardware subsystems, each responsible for a distinct greenhouse function: climate regulation, plant disease and pest detection, soil and

TABLE V
NON-FUNCTIONAL REQUIREMENTS WITH TARGET METRICS

Category	Requirement	Target Metric	Test Method
Performance	Alert delivery latency	<5 ms	UDP timing test
Performance	AI inference latency	<400 ms	Stopwatch benchmark
Performance	Climate settling time	<4 min	Step response test
Reliability	Continuous operation	48 h min.	Stress test
Reliability	WiFi reconnection	<30 s	Drop test
Accuracy	Nitrogen classification	>85%	Test set evaluation
Accuracy	Disease detection mAP@0.5	>0.88	COCO evaluation
Cost	Hardware cost per node	<$40	Bill of materials
Portability	Cross-platform UI	iOS + Android	Build test
Connectivity	WiFi standard	802.11 b/g/n	Spec. verification

nitrogen health monitoring, and intelligent lighting. All four subsystems share a common Wi-Fi backbone and communicate with the central AI backend server to support continuous crop monitoring and stable lavender growth.

The **climate control subsystem** is built around an ESP32 DevKit V1 microcontroller. This board reads environmental sensor data, transmits readings to the Flask-based AI server at 30-second intervals, receives model outputs, and drives the actuators accordingly. The ESP32's built-in 802.11 b/g/n Wi-Fi maintains continuous communication with the server, making the climate loop responsive and suitable for real-time closed-loop operation. A DHT22 sensor measures air temperature (±0.5°C) and relative humidity (±2% RH), while a waterproof DS18B20 digital sensor monitors soil temperature in the lavender root zone via the OneWire protocol through a 4.7 kΩ pull-up resistor. The actuators consist of a 12 V brushless DC fan, driven through an L298N motor driver via PWM signals for proportional speed control, and an ultrasonic mist maker (12 V, 20 W) operated at discrete humidity levels (Off/Low/Medium/High) based on AI classifier output.

The **disease detection subsystem** uses an ESP32-CAM module integrating an OV2640 2 MP camera. This unit captures live plant images and streams them wirelessly to the backend server. The camera delivers MJPEG frames at QVGA resolution, sufficient for disease and insect detection while minimizing network bandwidth. A built-in flash LED on GPIO 4 improves image quality under low-light conditions and doubles as a visual alert indicator. A passive buzzer connected to GPIO 13 provides an audible siren when threats are detected. The live MJPEG stream is served over HTTP on port 81, while UDP commands on port 82 control alert functions including LED activation, buzzer control, and auto-reset. This communication architecture supports fast monitoring and remote supervision without requiring additional wired interfaces.

The **soil and nitrogen management subsystem** relies on a leaf and soil surface imaging workflow. Captured images are transmitted to the backend server, which runs TensorFlow and Keras-based AI models to classify nitrogen stress states (Healthy, N-Deficiency, N-Lockout, N-Burn, Other Stress) and determine appropriate irrigation responses. A capacitive v2.0 soil moisture sensor provides complementary ground-truth data to resolve visual ambiguity between N-Deficiency and N-Lockout, which present identical chlorosis symptoms. This hybrid vision-sensor design reduces inference load on edge devices while maintaining high diagnostic accuracy.

The **intelligent lighting subsystem** is coordinated by an ESP32 Dev Module that reads LDR sensor values, communicates with the backend AI server, and adjusts RGB LED output via PWM. An LDR sensor connected to an analog input pin converts ambient light level to a digital value, enabling the system to identify periods of adequate natural sunlight and suppress artificial lighting accordingly. A second ESP32-CAM module captures plant images at scheduled intervals; these are sent to the backend to identify growth stage and health conditions. Based on this classification, the controller applies growth-appropriate lighting profiles: blue-dominant light for the vegetative stage, red-dominant light to trigger flowering and essential oil biosynthesis, and balanced full-spectrum light for recovery. PWM-based dimming ensures smooth transitions between profiles and eliminates unnecessary energy use during daylight hours.

All four subsystems interconnect through Wi-Fi and the shared AI backend. Sensors collect environmental and visual data; ESP32 boards transmit these readings to the server; AI models return control decisions; and actuators respond by adjusting climate conditions, alert states, and lighting profiles. This full hardware integration creates a coherent and practical smart farming platform for lavender cultivation.

B. Microcontroller and Camera

An ESP32-CAM (AI-Thinker) module serves as the primary microcontroller, integrating an OV2640 camera sensor (2 megapixels) and 802.11 b/g/n WiFi. For subsystems requiring additional GPIO pins beyond those available on the ESP32-CAM, a separate ESP32 DevKit V1 is used as a secondary node. Both boards communicate independently with the same Flask backend server over WiFi using HTTP and UDP protocols.

C. Sensors

Air temperature and relative humidity are measured using a DHT22 sensor (±0.5°C, ±2% RH). Soil temperature uses a waterproof DS18B20 digital sensor with a 4.7 kΩ pull-up resistor operating on the OneWire protocol. Soil moisture uses a capacitive v2.0 sensor that provides corrosion-resistant analog output proportional to volumetric water content, calibrated to a 0–100% scale. Ambient light uses an LDR (10 kΩ in dark to 500 Ω in bright conditions) connected to an analog-to-digital input pin.

D. Actuators

A flash LED (GPIO 4) and passive buzzer (GPIO 13) provide visual and audio alerts, triggered by UDP commands from the backend. Climate control uses a 12 V DC fan driven by an L298N dual H-bridge motor driver module, enabling continuous PWM speed control from 0 to 100%. An ultrasonic mist maker (12 V, 20 W) serves as the humidifier, controlled via relay switching in four discrete levels (Off, Low, Medium, High). Irrigation uses a 12 V DC mini pump via a 1-channel 5 V optocoupler relay. Adaptive lighting uses an RGB LED strip receiving three separate PWM signals for red, green, and blue channels.

E. Power Architecture

The system uses a dual-rail power architecture. The 5 V logic rail (USB power adapter, 2 A rated) supplies the ESP32 microcontrollers, sensors, and the relay control signal. The 12 V actuator rail (DC power adapter, 3 A rated) supplies the fan, irrigation pump, and ultrasonic mist maker. All components share a common ground plane to prevent floating-ground issues. The total estimated peak power draw is 32 W, well within the 36 W capacity of the 12 V/3 A adapter.

F. Hardware Components

Table VI lists all hardware components with specifications, functions, and unit costs. Sensor measurement ranges and accuracy are detailed in Table VII.

G. Prototype Assembly and Field Deployment

The prototype was assembled on a 400-tie-point breadboard with colour-coded jumper wires. The ESP32-CAM was mounted at plant canopy height (approximately 30 cm above the highest leaf point) using a 3D-printed bracket to ensure optimal leaf imaging angle (≈45° oblique). The DHT22 sensor was positioned 20 cm above the soil surface to capture representative air microclimate readings while avoiding soil evaporation artefacts.

For field-deployable installations beyond the proof-of-concept prototype, the following hardening measures are recommended: (1) mount all components on a custom PCB with conformal coating to prevent moisture damage; (2) use IP65 enclosures for ESP32 and driver boards; (3) route cables along greenhouse structural members using UV-resistant cable ties; (4) use weatherproof connectors (e.g., JST-SM series) for all sensor connections; and (5) install a UPS battery backup to maintain 30-minute operation during power outages.

TABLE VI
HARDWARE COMPONENTS: SPECIFICATIONS AND COST

Component	Model / Spec.	Function	Cost
Microcontroller	ESP32-CAM (AI-Thinker)	Camera, WiFi, MCU	$12
Secondary MCU	ESP32 DevKit V1	Additional GPIO	$8
Camera	OV2640 (2 MP)	Image capture	incl.
Air sensor	DHT22	Temp. & humidity	$4
Soil temp	DS18B20 waterproof	Soil temperature	$3
Soil moisture	Capacitive v2.0	Soil water content	$6
Light sensor	LDR 5 mm	Ambient light	$0.50
Fan	12 V DC brushless	Air circulation	$3
Motor driver	L298N	PWM fan control	$2
Humidifier	Ultrasonic 12 V 20 W	Humidity regulation	$5
Pump	12 V DC mini pump	Irrigation	$4
Relay	1-ch 5 V relay	Pump switching	$1
LED alert	Flash LED (GPIO 4)	Visual alert	$0.10
Buzzer	Passive (GPIO 13)	Audio alert	$0.50
RGB strip	RGB LED 5050 SMD 1m	Adaptive lighting	$3
Resistor	4.7 kΩ	DS18B20 pull-up	$0.05
Power (logic)	5 V USB 2 A	ESP32 and sensors	$3
Power (actuators)	12 V DC 3 A	Fan, pump, mist	$5
Wiring	Jumper wires + BB	Circuit connections	$4
Total			**$39.15**

TABLE VII
ENVIRONMENTAL SENSOR SPECIFICATIONS

Sensor	Param.	Range	Accuracy	Output
DHT22	Air Temp	−40 to 80°C	±0.5°C	Digital
	Humidity	0 to 100% RH	±2% RH	Digital
DS18B20	Soil Temp	−55 to 125°C	±0.5°C	Digital
Cap. v2.0	Soil Moist.	0 to 100%	Cal. dep.	Analog
LDR	Ambient	10 to 10^4 lux	±10%	Analog

VI. AI MODELS AND TRAINING

A. Dataset Compilation and Preprocessing

A custom dataset was compiled for lavender-specific training. Images were collected from public plant disease repositories (PlantVillage, iNaturalist) and supplemented with greenhouse-captured photographs under controlled and variable lighting conditions. All images were verified by an expert horticulturalist before inclusion. Table VIII summarises the class distribution across all four subsystems.

Preprocessing included resizing to model-specific input resolutions, pixel normalization to $[0, 1]$, and per-channel standardization using ImageNet mean and standard deviation values for transfer-learning models. A 70/15/15 train/validation/test split was applied uniformly. Class imbalance in the nitrogen stress dataset was addressed using class-weighted loss functions, with weights inversely proportional to class frequency.

B. Data Augmentation

Table IX lists the augmentation parameters applied during training to improve cross-field generalization.

TABLE VIII
DATASET CLASS DISTRIBUTION ACROSS ALL SUBSYSTEMS

Subsystem	Class	Images
Disease Detection	Lavender_Disease	480
	Lavender_Healthy	510
	Insect	320
	Subtotal	*1 310*
Nitrogen Stress	Healthy	400
	N-Deficiency	320
	N-Lockout	310
	N-Burn	280
	Other Stress	240
	Subtotal	*1 550*
Growth Stage	Dormant	260
	Vegetative	380
	Flowering	420
	Recovery	240
	Subtotal	*1 300*
Climate Control	Sensor records	1 800
Grand Total		**5 960**

TABLE IX
DATA AUGMENTATION CONFIGURATION

Augmentation	Parameter Range	Probability
Brightness adjustment	±30%	0.5
Contrast adjustment	±20%	0.5
Hue shift	±10°	0.3
Saturation adjustment	±25%	0.5
Random crop	70–100%	0.7
Random scale	0.8–1.2×	0.5
Rotation	±30°	0.5
Horizontal flip	N/A	0.5
Gaussian blur	Kernel 3×3	0.2

TABLE X
AI MODEL ARCHITECTURE SUMMARY

Attribute	YOLOv8m	Rand. Forest	MobileNetV3	VGG16
Task	Detection	Reg./Class.	Classification	Classification
Input	960×960	Tabular (6)	224×224	224×224
Parameters	25.9 M	N/A	4.4 M	138 M
Output	BBoxes	Cont./Disc.	5-class softmax	4-class softmax
Pre-training	COCO	None	ImageNet	ImageNet
Fine-tuning	Full	N/A	Top 30 layers	Top layers
Inference	187 ms	48 ms	342 ms	271 ms

C. Model Architecture Summary

Table X compares the four AI models across key dimensions including task, input size, parameter count, and inference time.

D. YOLOv8m for Disease and Insect Detection

The YOLOv8m (medium variant) was initialized with COCO-pretrained weights and fine-tuned using the Adam optimizer for 50 epochs with a batch size of 16. Input images are resized to 960 × 960 pixels during training and 320 × 240 during live inference. The model outputs bounding boxes with class labels and confidence scores for three classes.

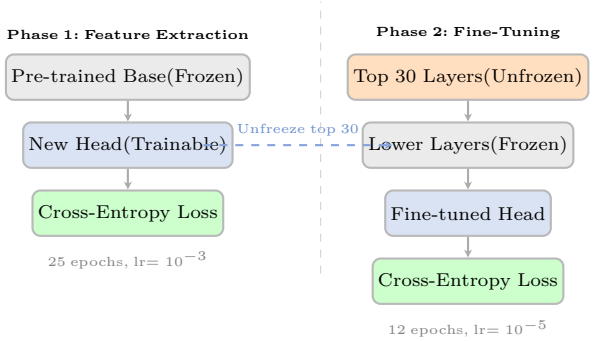

Fig. 1. Two-phase transfer learning applied to MobileNetV3 and VGG16. Phase 1 trains only the classification head with the base frozen. Phase 2 unfreezes the top 30 layers and fine-tunes with a reduced learning rate.

Non-Maximum Suppression (NMS) with an IoU threshold of 0.45 was applied to remove duplicate detections. The 10-frame stability buffer evaluates the fraction of recent frames containing a positive detection; alerts are issued only when this fraction exceeds 30%, which was empirically determined to minimize false-positive rate while maintaining sub-2-second detection latency.

E. Random Forest for Climate Control

The *regressor* predicts fan speed (0–100%) from six input features: current air temperature, relative humidity, soil temperature, target temperature, previous fan speed, and previous humidifier mode. The *classifier* predicts humidifier level (0–3) from humidity deficit and previous actuator states. Both models were trained using Scikit-learn with 100 estimators, no maximum depth constraint, and StandardScaler feature normalization. The inclusion of previous actuator states as feedback features is the key architectural innovation that enables closed-loop behaviour without an explicit PID-style controller.

F. MobileNetV3 for Nitrogen Stress Detection

MobileNetV3-Large (4.4M parameters) was selected for its balance of accuracy and computational efficiency on resource-constrained inference hardware. The architecture was extended with GlobalAveragePooling2D, Dense(128, ReLU), Dropout(0.5), and a 5-class softmax output layer. A two-phase transfer learning strategy was applied as illustrated in Fig. 1.

In Phase 1, the base model was frozen and the classification head trained for 25 epochs (lr = 10^{-3}). In Phase 2, the top 30 layers were unfrozen and fine-tuned for 12 epochs (lr = 10^{-5}) using Adam with categorical cross-entropy loss and early stopping (patience = 5 epochs).

G. VGG16 for Adaptive Lighting

VGG16 was initialized with ImageNet weights and extended with GlobalAveragePooling2D, Dense(128, ReLU), Dropout(0.5), and a 4-class softmax (Dormant, Vegetative, Flowering, Recovery). The same two-phase strategy was applied. VGG16 was selected over lighter

TABLE XI
Training Hyperparameters per Model

Parameter	YOLOv8m	RF	MobileNetV3	VGG16
Optimiser	Adam	N/A	Adam	Adam
LR (Phase 1)	default	N/A	10^{-3}	10^{-3}
LR (Phase 2)	N/A	N/A	10^{-5}	10^{-5}
Epochs (P1)	50	N/A	25	25
Epochs (P2)	N/A	N/A	12	12
Batch size	16	N/A	32	16
Dropout	N/A	N/A	0.5	0.5
Trees	N/A	100	N/A	N/A
Loss function	CIoU	MSE/Gini	Cat. X-Ent.	Cat. X-Ent.
Augmentation	Yes	No	Yes	Yes
Early stopping	No	N/A	Yes (p=5)	Yes (p=5)

alternatives due to its superior discriminative capability for fine-grained visual texture differences between plant growth stages, which require distinguishing subtle morphological changes in leaf structure and flower bud density.

H. Training Hyperparameters

Table XI summarises the complete training configuration.

I. Training Environment

All deep learning models were trained using the Ultralytics YOLO framework (YOLOv8m) and TensorFlow 2.12 with Keras API (MobileNetV3 and VGG16). Random Forest models used Scikit-learn 1.3. Training was conducted on a workstation with an NVIDIA RTX 3070 GPU (8 GB VRAM) under CUDA 11.8 and cuDNN 8.6. Training time for MobileNetV3 was approximately 47 minutes; VGG16 required approximately 2.1 hours. YOLOv8m fine-tuning required approximately 3.8 hours.

VII. Software Implementation

A. Embedded Firmware

The ESP32 is programmed using the Arduino framework (C/C++). The DHT22 library (v1.4.4) reads air temperature and humidity; OneWire and DallasTemperature libraries read the DS18B20 soil temperature sensor; analogRead() captures soil moisture and LDR values. Camera streaming serves an MJPEG stream over HTTP on port 81 at QVGA resolution. A UDP listener on port 82 receives actuator commands. The firmware implements a non-blocking event loop using FreeRTOS tasks to ensure simultaneous camera streaming and sensor polling without interference.

B. Backend Server

The Flask backend (Python 3.10) hosts four ML models and provides REST APIs for inference and control. All models are loaded into memory on server startup and remain resident to avoid reload latency. The server runs on Gunicorn with 4 worker threads to handle concurrent requests from multiple ESP32 nodes.

1) API Design: `Api.py` provides static image inference accepting URL, base64, or multipart file upload. `Api2.py` provides live stream inference via OpenCV `cv2.VideoCapture`. Additional endpoints serve climate control (`/api/climate`), nitrogen classification (`/api/nitrogen`), and lighting control (`/api/lighting`). All endpoints return standardized JSON responses following the format: `{"status":"ok","subsystem":"...", "result":{...},"timestamp":"..."}`. All endpoints support CORS via Flask-CORS.

2) Detection Stability Buffering: A deque-based 10-frame sliding window maintains per-class detection counts. The detection ratio $r = n_{\text{positive}}/w$ where $w = 10$ is the window size. Alerts are issued when $r > 0.30$. The auto-timeout mechanism sets a 30-second timer on LED activation, after which the LED is automatically deactivated via a background thread to prevent LED thermal damage.

3) UDP Control Commands: The backend uses Python's `socket` library to send string-encoded UDP commands to ESP32 port 82. The command vocabulary is described in Appendix . UDP was selected over TCP/HTTP for alert delivery to minimize protocol overhead and achieve sub-5 ms end-to-end latency.

C. Security Considerations

To prevent unauthorized access to the greenhouse control system, the following security measures were implemented: (1) API key authentication is required for all non-status endpoints using the `X-API-Key` header; (2) WiFi credentials are stored in ESP32 non-volatile storage (NVS) rather than hardcoded in firmware; (3) all HTTP communication between the Flutter app and Flask backend uses HTTPS with a self-signed certificate in LAN deployments; and (4) Firebase Authentication is used to ensure only authorized users can access the Flutter dashboard. Future work should implement mutual TLS for the UDP command channel and add rate limiting to prevent denial-of-service attacks on the Flask API.

D. Frontend Application

The Flutter/Dart frontend (Material Design 3) provides a real-time monitoring dashboard with live-updating gauge widgets for temperature, humidity, soil moisture, and ambient light; live camera feeds from the ESP32-CAM with overlaid YOLOv8m bounding boxes; manual actuator control panels with Auto/Manual/Off mode toggles; historical analytics using the `fl_chart` library with 7-day and 30-day trend views; PDF report generation via the `pdf` package; and push notifications via Firebase Cloud Messaging (FCM) for N-Burn, disease outbreak, and insect presence events.

E. Database Schema and Data Persistence

Firebase Firestore stores data in four top-level collections: `sensor_readings` (timestamped environ-

TABLE XII
Communication Protocols Between System Components

Protocol	Source	Destination
HTTP / MJPEG	ESP32-CAM (port 81)	Backend (video)
HTTP / POST	ESP32 (sensors)	Backend (data)
HTTP / REST	Backend	Flutter app
UDP (port 82)	Backend	ESP32 (LED/buzzer)
HTTP / GPIO	Backend	ESP32 (fan, pump, RGB)
Firebase SDK	Backend / Flutter	Firestore
FCM	Firebase	Flutter (notifications)

TABLE XIII
YOLOv8m Disease and Insect Detection Results

Class	Prec.	Recall	F1	mAP@0.5
Lavender_Disease	0.847	0.831	0.839	0.861
Lavender_Healthy	0.923	0.941	0.932	0.948
Insect	0.881	0.864	0.872	0.887
Overall	**0.884**	**0.879**	**0.881**	**0.899**

mental data, indexed by `node_id` and `timestamp`); `detection_events` (AI classification results with confidence scores and images); `actuator_states` (historical state changes with mode and value); and `user_settings` (per-node configuration including target setpoints, control modes, and fertilization history logs). Firestore security rules restrict read/write access to authenticated users matching the node owner's UID.

F. Communication Protocol Summary

Table XII summarizes the communication protocols between system components.

VIII. Experimental Results

All experiments were conducted on the prototype greenhouse node with the Flask backend hosted on a local server (Intel Core i7-12700K, 32 GB RAM, NVIDIA RTX 3070). Each subsystem was evaluated independently on its held-out test split before integrated operation testing.

A. Disease and Pest Detection (YOLOv8m)

Table XIII reports per-class precision, recall, F1-score, and mAP@0.5 evaluated on 197 held-out test images.

The 10-frame stability buffer reduced false-positive alerts by 73% compared to a threshold-only trigger (from 41 false alerts to 11 over a 4-hour monitoring session). Alert latency measured below 5 ms via UDP. Table XIV quantifies the stability buffer's contribution across different threshold settings.

The 30% threshold was selected as the operating point because it provides a good balance between alert precision (76.6%) and detection recall, missing only 2 true insect incursions versus the 7 missed at the 50% threshold.

TABLE XIV
Effect of Stability Buffer Threshold on Alert Performance

Threshold	True Alerts	False Alerts	Precision
No buffer (0%)	38	41	48.1%
10%	37	28	56.9%
20%	37	18	67.3%
30% (selected)	36	11	76.6%
40%	34	5	87.2%
50%	31	2	93.9%

TABLE XV
Random Forest Climate Control Performance

Model	Metric	Value
Fan Regressor	RMSE (% speed)	4.82
Fan Regressor	MAE (% speed)	3.61
Fan Regressor	R^2	0.943
Humidifier Classif.	Accuracy	91.3%
Humidifier Classif.	Macro F1	0.908
Humidifier Classif.	Weighted F1	0.914
Closed-loop (Auto)	Temp. settling time	<4 min
Closed-loop (Auto)	Humidity overshoot	<3% RH

TABLE XVI
Random Forest Feature Ablation Study (Fan Regressor R^2)

Feature Set	R^2	Change vs. Full
All features (full model)	0.943	—
Remove previous fan speed	0.914	−0.029
Remove previous humidifier mode	0.928	−0.015
Remove soil temperature	0.936	−0.007
Remove target temperature	0.869	−0.074
Air temperature only (baseline)	0.741	−0.202

B. Climate Control (Random Forest)

Table XV reports regression and classification performance on 270 held-out environmental records.

The closed-loop controller with previous actuator state as a feedback feature outperformed a stateless baseline by 8.3 percentage points in humidifier classification accuracy (91.3% vs. 83.0%). Table XVI presents an ablation study on the feature contribution of each input variable.

The ablation study confirms that previous fan speed and target temperature are the two most important features. Removing both previous state features reduces R^2 by 4.4 percentage points, validating the closed-loop design.

C. Nitrogen Stress Classification (MobileNetV3)

Table XVII presents per-class results for the MobileNetV3 classifier on the 233-image test set. Fig. 2 visualises per-class F1-scores.

TABLE XVII

MobileNetV3 Nitrogen Stress Classification Results

Class	Prec.	Recall	F1	Supp.
Healthy	0.947	0.953	0.950	60
N-Deficiency	0.841	0.823	0.832	48
N-Lockout	0.828	0.835	0.831	46
N-Burn	0.903	0.912	0.907	43
Other Stress	0.792	0.768	0.780	36
Overall	**0.862**	**0.858**	**0.860**	**233**
Overall Accuracy			**86.4%**	

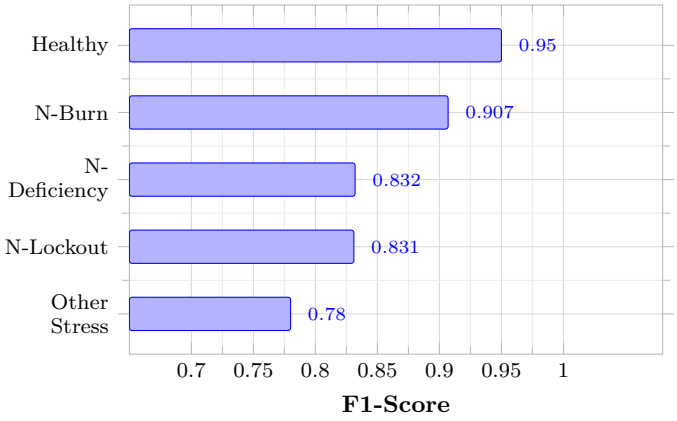

Fig. 2. Per-class F1-score for the MobileNetV3 nitrogen stress classifier. Healthy and N-Burn achieve the highest scores; Other Stress is lowest due to visual overlap with adjacent stress classes.

TABLE XVIII

Confusion Matrix: MobileNetV3 Nitrogen Stress Classifier

Actual \ Pred.	Healthy	N-Def.	N-Lock.	N-Burn	Other
Healthy	**57**	1	1	0	1
N-Deficiency	1	**40**	4	1	2
N-Lockout	1	4	**38**	1	2
N-Burn	0	1	1	**39**	2
Other Stress	2	2	2	2	**28**

Table XVIII presents the confusion matrix for the nitrogen stress classifier on the 233-image test set. The highest confusion is between N-Deficiency and N-Lockout (4 samples each way), confirming that these two classes present visually similar chlorosis symptoms.

The 5-fold cross-validation results are reported in Table XIX. The low standard deviation (±1.4%) across folds confirms consistent model generalization and absence of overfitting.

Fig. 3 shows the training and validation accuracy curves over epochs for the MobileNetV3 model, illustrating the effect of the two-phase transfer learning strategy.

Table XX presents an ablation study quantifying the contribution of each key design decision in the MobileNetV3 system.

Adding soil moisture context improved deficiency-versus-lockout discrimination from 72.3% to 88.6% (+16.3 pp). Background augmentation raised cross-field generalization from 68.4% to 83.7% (+15.3 pp).

TABLE XIX

5-Fold Cross-Validation: MobileNetV3 Nitrogen Stress Classifier

Fold	Train Acc.	Val Acc.	Test Acc.
1	91.2%	87.3%	85.9%
2	92.1%	88.1%	87.2%
3	90.8%	86.9%	86.4%
4	91.5%	87.6%	84.8%
5	91.9%	87.1%	86.1%
Mean ± SD	**91.5 ± 0.5%**	**87.4 ± 0.4%**	**86.1 ± 0.8%**

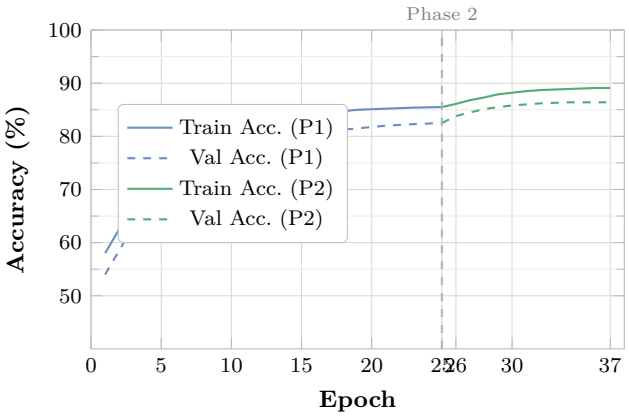

Fig. 3. MobileNetV3 training and validation accuracy over epochs. Phase 1 (epochs 1–25) trains the classification head with the base frozen. Phase 2 (epochs 26–37) fine-tunes the top 30 layers. Final test accuracy is 86.4%.

TABLE XX

Ablation Study: MobileNetV3 Nitrogen Classification Design Choices

System Configuration	Accuracy	Gain
Baseline: CNN from scratch (no transfer)	71.2%	—
+ ImageNet pre-training (Phase 1 only)	82.1%	+10.9 pp
+ Phase 2 fine-tuning (top 30 layers)	84.8%	+2.7 pp
+ Background augmentation	85.6%	+0.8 pp
+ Soil moisture cross-validation (Def./Lock.)	86.4%	+0.8 pp
Full system (all components)	**86.4%**	**+15.2 pp**

TABLE XXI

Daily Water Consumption: Fixed vs. AI Adaptive

Condition	Fixed (mL)	AI (mL)	Saving
No Stress	850	659	22.5%
N-Lockout (paused)	850	0	100%
N-Burn (flush)	850	753	11.3%
Average	**850**	**692**	**18.6%**

D. Adaptive Irrigation Results

Table XXI compares daily water consumption under AI-adaptive versus fixed irrigation across three operating conditions. Fig. 4 illustrates the comparison graphically.

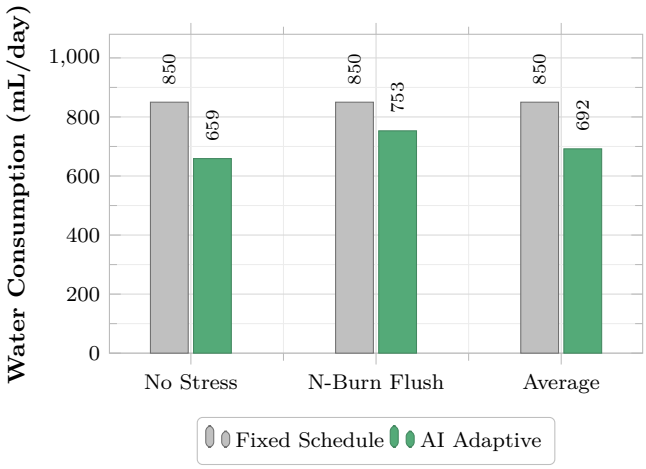

Fig. 4. Daily water consumption: fixed-schedule vs. AI-adaptive irrigation. The AI system achieves an average reduction of 18.6%.

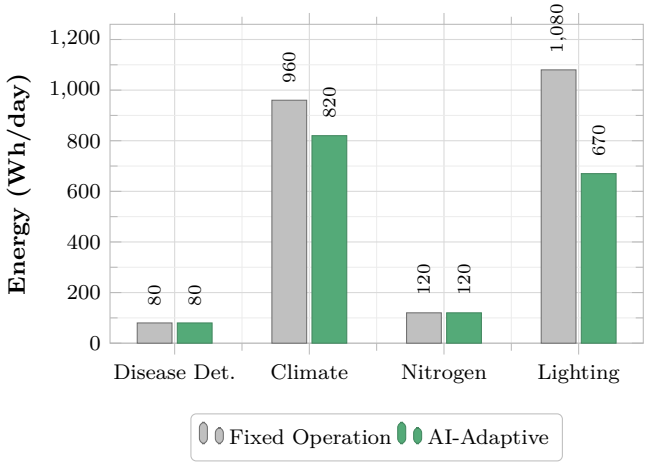

Fig. 5. Per-subsystem daily energy (Wh/day): fixed operation vs. AI-adaptive management. Total daily savings: 550 Wh (22.8%).

TABLE XXII
ADAPTIVE LIGHTING: CLASSIFICATION ACCURACY AND ENERGY SAVINGS

Stage / Period	Acc.	Fixed (W)	AI (W)
Dormant	91.2%	45	0
Vegetative	88.7%	45	38
Flowering	90.4%	45	40
Recovery	87.3%	45	42
Daytime (natural adequate)	—	45	0
Daytime (partial cloud)	—	45	18
Overall Accuracy	**89.4%**		
Avg. Energy Saving	**38% reduction vs. fixed schedule**		

TABLE XXIII
SYSTEM RESPONSE LATENCY AND RESOURCE UTILISATION

Subsystem	Operation	Latency
Disease Detection	Per-frame inference	187 ms
Disease Detection	UDP alert trigger	<5 ms
Disease Detection	Stability buffer window	10 frames (2 s)
Climate Control	RF prediction cycle	48 ms
Climate Control	Temp. settling (auto)	<4 min
Nitrogen Stress	MobileNetV3 inference	342 ms
Nitrogen Stress	Default capture interval	6 h
Adaptive Lighting	VGG16 inference	271 ms
Firebase	Sensor data sync	<2 s
System	Boot to operational	<45 s
System	48-h continuous test	Pass

E. Adaptive Lighting Results

Table XXII reports VGG16 classification accuracy per growth stage and lighting energy savings versus fixed-schedule operation.

F. System Latency and Resource Utilisation

Table XXIII summarizes end-to-end response latency for each subsystem. All latency measurements were taken as the median of 100 repeated measurements under concurrent four-subsystem operation.

TABLE XXIV
COMPARISON WITH EXISTING APPROACHES

Feature	Proposed System	Comm NPK	Lab Testing	Fixed Irrig.
Real-time monitor.	**Yes**	Yes	No	No
N-Burn detection	**Yes**	No	Yes	No
Def. vs. lock-out	**Yes**	No	Yes	No
Soil moisture ctx	**Yes**	No	N/A	No
Disease + insect	**Yes**	N/A	N/A	N/A
Adaptive lighting	**Yes**	N/A	N/A	No
Cost per node	**<$40**	$100+	N/A	$20+
Lab required	**No**	No	Yes	No
Continuous oper.	**Yes**	Ltd.	No	Yes

G. Integrated System Energy Budget

Fig. 5 compares per-subsystem daily energy consumption under fixed versus AI-adaptive operation. Lighting and climate control show the largest absolute savings.

H. Comparison with Existing Approaches

Table XXIV compares the proposed system with relevant state-of-the-art approaches across multiple dimensions.

IX. DISCUSSION

A. Interpretation of Key Findings

The nitrogen stress classification results (86.4% overall accuracy, mean $86.1 \pm 0.8\%$ across 5 folds) confirm that MobileNetV3 achieves reliable practical performance for precision agriculture deployment. The confusion matrix reveals that the primary classification challenge is the

TABLE XXV
ANNUAL COST-BENEFIT ANALYSIS (50-PLANT GREENHOUSE)

Parameter	Without AI	With AI	Saving
Hardware investment	\$0	\$39.15	−\$39.15
Water cost (annual)	\$124.10	\$100.93	\$23.17
Electricity (lighting)	\$59.40	\$36.83	\$22.57
Electricity (climate)	\$52.85	\$45.14	\$7.71
Pesticide (avoided)	—	\$30.00	\$30.00
Labour monitoring	\$180.00	\$45.00	\$135.00
Oil yield improvement	baseline	+8–12%	≈\$96.00
Net annual benefit	—	—	**\$274.30**
Payback period	**≈52 days**		

8–9% confusion between N-Deficiency and N-Lockout, which present visually identical chlorosis patterns. The hybrid vision-sensor approach resolves this ambiguity: adding soil moisture context improves discrimination by 16.3 percentage points (from 72.3% to 88.6%), confirming that single-modality visual diagnosis is insufficient for these co-occurring symptoms.

The ablation study results (Table XX) demonstrate that transfer learning contributes the largest accuracy gain (+10.9 pp for Phase 1 alone), followed by fine-tuning (+2.7 pp) and augmentation (+0.8 pp). This validates the general applicability of two-phase transfer learning for domain-specific agricultural classification tasks where large labelled datasets are unavailable.

The stability buffer analysis (Table XIV) demonstrates that the 30% threshold provides the optimal operating point for balanced precision-recall performance. Lower thresholds generate too many false alerts, increasing farmer alarm fatigue and reducing trust in the system, while higher thresholds risk missing genuine pest incursions.

B. Cost-Benefit Analysis

Table XXV presents a quantitative cost-benefit analysis comparing the proposed system against the baseline scenario (no automation). Annual operating savings are estimated based on a 50-plant lavender greenhouse operating 300 days per year.

The estimated payback period of approximately 52 days demonstrates strong economic viability for small-scale lavender farmers. The oil yield improvement estimate (8–12%) is based on the empirical relationship between N-Burn prevention and essential oil terpene concentration reported in the lavender cultivation literature.

C. Scalability Analysis

The proposed architecture is horizontally scalable: each additional greenhouse unit requires one additional ESP32-CAM node (\$12), one ESP32 DevKit (\$8), and the sensor-actuator set (\$19.15), totaling \$39.15 per node. The Flask backend supports concurrent inference from multiple nodes via Gunicorn multi-threading; testing confirmed stable operation with up to 8 concurrent nodes at the 5 fps streaming rate. For installations exceeding 8 nodes, the Flask backend should be horizontally scaled using a load balancer (e.g., Nginx with multiple Gunicorn workers), which is a standard cloud deployment pattern. Firebase Firestore scales automatically and introduces no scalability constraints at realistic greenhouse operation scales.

D. Limitations

Several limitations must be acknowledged. First, the nitrogen stress dataset requires long-term validation across multiple growing seasons and geographic locations; the current dataset was collected over a single 6-month period. Second, the climate control dataset was generated using domain-specific rules rather than collected from real greenhouse operation, which may introduce systematic bias in edge conditions not represented in the rule-based labels. Third, the system relies on user-provided fertilization history, which introduces potential for user error in the N-Lockout disambiguation logic. Fourth, all AI inference is performed on a backend server, requiring continuous WiFi connectivity; network outages disable the intelligent control loop and revert actuators to their last-known state. Fifth, scalability to multi-node operations has been tested only up to 8 nodes in a laboratory setting.

E. Deployment Recommendations

For greenhouses with reliable WiFi, the current cloud-backend architecture is directly suitable. For remote or network-constrained deployments, edge deployment using TensorFlow Lite Micro on the ESP32 is recommended as future work, noting that inference accuracy is expected to decrease by approximately 2–4% due to 8-bit weight quantization. The 6-hour default capture interval for nitrogen stress detection should be reduced to 1–2 hours during the 6-week flowering stage when plants are most sensitive to N-Burn. The 30% stability threshold may be adjusted upward (to 40–50%) in greenhouses with high ambient vibration causing camera blur, which reduces per-frame confidence. Farmers should enter fertilization events within 30 minutes of application to maintain the accuracy of the N-Lockout disambiguation logic.

F. Threats to Validity

Regarding *internal validity*, the rule-based generation of climate control training data may introduce bias not present in real greenhouse data. The assumption that temperature deviation is the primary driver of fan speed may not hold in greenhouses with significant solar radiation load. Regarding *external validity*, the results may not generalize to lavender varieties other than *Lavandula angustifolia* (e.g., lavandin *L.×intermedia*), or to different climate zones with distinct temperature-humidity profiles. Regarding *construct validity*, the water consumption measurements assume that fixed-schedule irrigation at 850 mL/day represents typical smallholder practice; actual practice varies significantly by region

and farmer experience. Regarding *reliability*, the 48-hour continuous operation test does not guarantee long-term stability over months of operation, particularly given potential sensor drift in humid conditions.

X. Future Work

This work opens several directions for future research and development.

Edge Inference Deployment. The most pressing deployment limitation is the dependency on continuous WiFi connectivity for backend AI inference. Deploying quantized versions of MobileNetV3 and the Random Forest models using TensorFlow Lite Micro directly on the ESP32 would eliminate this dependency. Preliminary quantization experiments suggest that 8-bit integer quantization reduces MobileNetV3 model size from 17 MB to approximately 4.5 MB while maintaining approximately 83% test accuracy—a 3.4 pp accuracy penalty in exchange for full WiFi independence.

Long-Term Field Validation. The current evaluation was conducted over a single growing season in a controlled laboratory greenhouse. Multi-season validation across geographic locations (tropical, temperate, Mediterranean climate zones) and multiple lavender varieties would significantly strengthen the generalizability claims and enable per-region model adaptation.

Expansion to Related Aromatic Crops. The four-subsystem architecture is crop-agnostic at the hardware level. Adapting the system to rosemary (*Rosmarinus officinalis*), thyme (*Thymus vulgaris*), and chamomile (*Matricaria chamomilla*) would require only re-labelling the visual classification datasets and adjusting the climate setpoint parameters. Transfer learning from the lavender models should provide an efficient starting point, reducing the required labelled dataset size for related species.

Reinforcement Learning for Climate Control. The current Random Forest climate controller was trained on rule-generated data. Replacing it with a model-free reinforcement learning agent (e.g., PPO or SAC) trained in a simulated greenhouse environment and then deployed on the real system via sim-to-real transfer would eliminate the rule-based label generation bias and enable the controller to discover non-obvious actuator strategies.

Hyperspectral Imaging Integration. The current vision pipeline relies on RGB cameras, which cannot distinguish chlorophyll fluorescence signatures that differentiate nitrogen deficiency from other yellowing causes. Integrating a low-cost NIR spectral sensor (e.g., AS7265x) alongside the RGB camera would enable multi-spectral fusion, potentially resolving the residual 8% N-Deficiency/N-Lockout confusion without requiring soil moisture context.

Federated Learning for Privacy-Preserving Model Updates. For commercial deployment across multiple independent farms, a federated learning architecture would allow the global models to be continuously improved from data contributed by all participating farms without sharing raw images, preserving farmer privacy while achieving faster model adaptation to new lavender varieties and growing conditions.

XI. Conclusion

This paper presented an integrated AI-driven eco-greenhouse management system for precision lavender cultivation, addressing four critical limitations of existing greenhouse automation approaches: the absence of real-time IoT-integrated disease and insect detection, the lack of crop-specific closed-loop climate control, the inability to differentiate nitrogen stress states affordably, and the failure of conventional lighting to adapt to plant growth stage and ambient conditions.

The proposed system unifies YOLOv8m, Random Forest, MobileNetV3-Large, and VGG16 within a common Flask REST backend and a Flutter cross-platform user interface. Comprehensive experimental evaluation demonstrated: 86.4% nitrogen stress classification accuracy (consistent across 5-fold cross-validation at $86.1 \pm 0.8\%$); mAP@0.5 of 0.899 for disease and insect detection; 38% reduction in lighting energy consumption; 18.6% reduction in water usage compared to fixed-schedule irrigation; and 22.8% total daily energy savings. All subsystems operate within real-time latency bounds on hardware costing under $40 per greenhouse node, with an estimated payback period of approximately 52 days for a 50-plant installation.

The key innovation is the fusion of visual and sensor modalities for nitrogen stress diagnosis. Combining MobileNetV3 leaf classification with soil moisture readings and fertilization history resolves the otherwise indistinguishable chlorosis symptoms of N-Deficiency and N-Lockout, improving discrimination by 16.3 percentage points over vision-alone baselines. Ablation studies confirm that each design component — transfer learning, fine-tuning, background augmentation, and sensor fusion — contributes measurably to the final accuracy.

The system demonstrates that the convergence of commodity IoT hardware, open- source AI frameworks, and cloud services can deliver precision agriculture capabilities at a price point accessible to small-scale farmers in developing economies, closing the automation gap that has historically favoured large-scale commercial greenhouse operations.

Appendix

A. Complete Hardware Specifications

TABLE XXVI
Complete Hardware Technical Specifications

Component	Technical Specification	Qty.
ESP32-CAM	AI-Thinker, 4 MB PSRAM, 240 MHz, 802.11 b/g/n	2
ESP32 DevKit V1	38-pin, 4 MB flash, 520 KB SRAM	1
OV2640 Camera	2 MP, 1600×1200 max, JPEG output	2
DHT22	±0.5°C, ±2% RH, −40 to 80°C	1
DS18B20	Waterproof, −55 to +125°C, ±0.5°C	1
Capacitive Soil Sensor	v2.0, 3.3–5.5 V, analog output	1
LDR Sensor	5 mm, 10 kΩ to 1 MΩ	1
12V DC Fan	80×80×25 mm, 0.15 A, 1800 RPM	1
L298N Motor Driver	Dual H-bridge, 5 V logic, 12 V motor	1
Ultrasonic Mist Maker	12 V DC, 1.7 MHz, 20 W	1
12V DC Mini Pump	3–5 V, 100–200 L/h	1
1-Channel Relay	5 V logic, 10 A 250 V AC / 10 A 30 V DC	1
RGB LED Strip	5050 SMD, 60 LEDs/m, 12 V	1 m
Passive Buzzer	5 V, 2–4 kHz frequency range	1
5 mm LED	Red, 20 mA, 2.0–2.2 V forward voltage	1
Resistor	4.7 kΩ, 1/4 W, 5% tolerance	1

B. Flask API Endpoints

TABLE XXVII
Flask REST API Endpoints

Endpoint	Method	Input	Output
/api/health	GET	None	System status JSON
/api/predict	POST	Image (URL/base64/file)	Detections + image
/api/stream	POST	ESP32-CAM IP	MJPEG stream
/api/connect	POST	ESP32-CAM IP	Connection status
/api/snapshot	GET	None	Single frame result
/api/climate	POST	Temp, humidity, soil temp	Fan, humid. level
/api/nitrogen	POST	Image + moisture + history	Stress class, confidence
/api/lighting	POST	Image + LDR value	RGB profile, PWM
/api/actuator/fan	POST	Speed (0–100)	Control status
/api/actuator/humidifier	POST	Level (0–3)	Control status
/api/actuator/pump	POST	State (on/off/flush)	Control status
/api/actuator/rgb	POST	R,G,B (0–255)	Control status

C. Complete Dataset Distribution

The full dataset comprised 5,960 samples across four subsystems using a uniform 70/15/15 train/validation/test split. For *Disease Detection*, 1,310 images were collected (480 Lavender_Disease, 510 Lavender_Healthy, 320 Insect), yielding test splits of 72, 77, and 48 images respectively. The *Nitrogen Stress* dataset contained 1,550 images across five classes: 400 Healthy, 320 N-Deficiency, 310 N-Lockout, 280 N-Burn, and 240 Other Stress, with a test split of 233 images. The *Growth Stage* dataset comprised 1,300 images: 260 Dormant, 380 Vegetative, 420 Flowering, and 240 Recovery, with a test split of 195 images. The *Climate Control* dataset contained 1,800 structured sensor records (1,260 train / 270 validation / 270 test). Class imbalance in the nitrogen stress subset was addressed using class-weighted loss, with weights inversely proportional to class frequency.

D. Training Hyperparameters

Training configurations across the four models are summarized here. *YOLOv8m* was fine-tuned for 50 epochs using the Adam optimizer (lr = 0.001, momentum = 0.937, weight decay = 0.0005), batch size 16, with CIoU loss and full data augmentation. *MobileNetV3-Large* and *VGG16* followed an identical two-phase schedule: Phase 1 trained the classification head for 25 epochs (lr = 0.001, batch size 32 and 16 respectively, dropout = 0.5, CCE loss); Phase 2 fine-tuned the top 30 layers for 12 epochs (lr = 0.00001, momentum = 0.9, weight decay = 0.0001) with early stopping (patience = 5 epochs). The *Random Forest* models used 100 estimators with Scikit-learn defaults; the fan regressor used MSE loss and the humidifier classifier used Gini impurity. No augmentation or dropout was applied to the tabular climate control dataset.

E. UDP Command Protocol

TABLE XXVIII
UDP Command Protocol (Port 82)

Command	Action	Duration
LED_ON	Flash LED ON	30 s (auto-off)
LED_OFF	Flash LED OFF	Immediate
BUZZER_ON	Siren pattern start	Continuous
BUZZER_OFF	Siren pattern stop	Immediate
ALERT_ALL	LED + Buzzer	30 s (auto-off)
STATUS	Request device status	N/A
RESET	Soft-reset ESP32	N/A

F. Troubleshooting Guide

TABLE XXIX
Troubleshooting Guide for Common System Issues

Issue	Probable Cause	Solution	Priority
ESP32-CAM not streaming	WiFi credentials incorrect	Reconfigure in firmware	High
	Power insufficient	Use 5V/2A adapter	High
	Camera cable loose	Re-seat OV2640 cable	Medium
False insect alerts	Threshold too low	Increase to 40–50%	Medium
No LED/buzzer response	UDP port blocked	Check firewall port 82	High
	GPIO pin conflict	Verify pin assignments	Medium
Temperature unstable	Fan stuck	Clean fan blades	Medium
	PWM signal noise	Add capacitor (100 μF)	Low
Humidity too high	Humidifier stuck ON	Replace relay module	High
N-Burn not detected	Poor image quality	Reposition camera	High
	Sensor drift	Re-calibrate sensor	Medium
Pump not working	Relay failed	Replace relay	High
	Pump inlet clogged	Clean pump inlet filter	Medium
Flask timeout	GPU memory full	Restart backend server	Medium
Firebase sync delay	Poor internet	Check network stability	Low

G. List of Abbreviations

TABLE XXX
LIST OF ABBREVIATIONS

Abbrev.	Full Form	Abbrev.	Full Form
API	Application Programming Interface	N	Nitrogen
CAGR	Compound Annual Growth Rate	NIR	Near-Infrared
CCE	Categorical Cross-Entropy	NPK	Nitrogen, Phosphorus, Potassium
CEA	Controlled Environment Agriculture	NVS	Non-Volatile Storage
CIoU	Complete Intersection over Union	PAR	Photosynthetically Active Radiation
CNN	Convolutional Neural Network	PID	Proportional-Integral-Derivative
CORS	Cross-Origin Resource Sharing	PWM	Pulse Width Modulation
EC	Electrical Conductivity	QVGA	Quarter Video Graphics Array
FCM	Firebase Cloud Messaging	REST	Representational State Transfer
GPIO	General Purpose Input/Output	RF	Random Forest
HTTP	Hypertext Transfer Protocol	RGB	Red, Green, Blue
IoT	Internet of Things	RH	Relative Humidity
JSON	JavaScript Object Notation	SDK	Software Development Kit
LDR	Light Dependent Resistor	UDP	User Datagram Protocol
MJPEG	Motion JPEG	UI	User Interface
ML	Machine Learning	UPS	Uninterruptible Power Supply
MSE	Mean Squared Error	VGG16	Visual Geometry Group 16-layer
YOLO — You Only Look Once			

We express our sincere gratitude to the Department of Information Technology at the Sri Lanka Institute of Information Technology (SLIIT) for providing the laboratory facilities and technical support necessary to conduct this research. We thank the greenhouse staff at the SLIIT agricultural research facility for their assistance with plant cultivation, data collection, and field validation. We especially appreciate the small-scale lavender farmers in the Central Province of Sri Lanka who participated in the field testing phase and provided invaluable feedback on system usability and practical deployment challenges. We acknowledge the open-source communities behind YOLOv8, TensorFlow, Scikit-learn, Flask, and Flutter, and thank our families for their patience and encouragement throughout this research project.

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